

RGB-VK Privileged Distillation for Privacy-Friendly Non-RGB Multimodal Human Sensing under Missing Modalities

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Abstract—Human sensing in smart spaces requires reliable perception of body structure and activity semantics while respecting privacy constraints during long-term deployment. Although RGB cameras provide rich appearance, contextual, and structural information, their continuous use raises privacy concerns and deployment limitations in residential and indoor environments. Non-RGB sensors such as depth, LiDAR, mmWave, and WiFi-CSI are more suitable for privacy-sensitive sensing, but their representations are often sparse, noisy, and vulnerable to missing sensor modalities. This paper presents an RGB-VK privileged distillation framework that treats RGB as training-time privileged information and uses visual keypoints (VK) as a privacy-friendly structured bridge between visual knowledge and non-RGB multimodal sensing. An RGB-VK Super-Teacher compresses visual appearance and context into transferable structural supervision, while a non-RGB multimodal Student learns to perform human pose estimation (HPE) and human activity recognition (HAR) without RGB input at inference. To improve deployment robustness, the Student is trained with randomly sampled modality subsets and reliability-calibrated fusion. Experiments on MMFi show that the proposed Student maintains strong performance under arbitrary modality combinations, achieving 75.18 mm average MPJPE over 31 HPE combinations and 85.15% average HAR accuracy over 15 modality combinations. These results demonstrate the potential of RGB-privileged training and non-RGB inference for privacy-friendly and missing-modality robust human sensing in smart spaces.

Index Terms—Human sensing, multimodal sensing, privacy-friendly perception, non-RGB sensing, visual keypoints, cross-modal distillation, missing modality, smart spaces, human pose estimation, human activity recognition.

I. Introduction

Human sensing is a fundamental capability for smart spaces, intelligent homes, embodied intelligence, assisted living, and human-computer interaction. In these scenarios, human pose estimation (HPE) provides fine-grained body structure, while human activity recognition (HAR) captures semantic behavior patterns. A practical sensing system should therefore support both geometric and semantic understanding, while remaining deployable in long-term indoor environments.

RGB cameras have been widely adopted for human perception because they provide dense appearance, contextual, and structural cues. However, continuous RGB

sensing exposes facial appearance, clothing, background, and personal activities, which raises privacy concerns in homes and other private spaces. RGB sensing is also affected by illumination, occlusion, and social acceptance. These issues make raw RGB unsuitable as the only inference modality for privacy-sensitive smart spaces.

Non-RGB sensing modalities, including depth, LiDAR, mmWave radar, and WiFi-CSI, provide alternative ways to perceive humans with reduced visual exposure. They are attractive for non-contact, low-intrusion, and privacy-friendly deployment. Nevertheless, they are also challenging: their signals are sparse, noisy, heterogeneous, and frequently incomplete in real deployments. A model trained under full-modality assumptions can degrade severely once a sensor fails or a subset of modalities becomes unavailable.

Existing multimodal fusion and missing-modality learning methods address parts of this problem, but they do not fully close the deployment gap between RGB-rich training and non-RGB inference. RGB-to-skeleton or pose distillation has also been studied, but it is usually not formulated as a structured privileged supervision bridge for non-RGB multimodal HPE/HAR under arbitrary sensor availability. Therefore, the key question is how to exploit RGB knowledge during training without requiring RGB during inference, while simultaneously improving robustness to missing non-RGB sensors.

This paper proposes an RGB-VK privileged distillation framework. RGB is used only as training-time privileged information, and visual keypoints (VK) are used as a privacy-friendly structured bridge that preserves human geometry while suppressing appearance-level information. An RGB-VK Super-Teacher transfers visual structural knowledge to a non-RGB multimodal Student. During inference, only non-RGB sensors are required, and the Student supports arbitrary available modality combinations through modality-subset training and reliability-calibrated fusion.

The main contributions of this paper are summarized as follows:

- We formulate VK as a privacy-friendly structured bridge that compresses RGB privileged knowledge into transferable body-structure supervision for non-RGB multimodal human sensing.
- We construct a non-RGB multimodal Student for HPE and HAR that does not require RGB during

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inference and is trained to operate under arbitrary missing sensor modalities.

- We introduce reliability-calibrated modality-wise prediction and fusion for robust HPE/HAR, and validate the framework on MMFi with 31 HPE modality combinations and 15 HAR modality combinations.

II. Related Work

A. Multimodal Human Sensing in Smart Spaces

Multimodal human sensing has been studied with RGB, depth, LiDAR, radar, WiFi, and CSI-based signals. These sensors provide complementary observations for human pose, motion, and activity understanding. RGB offers dense visual cues, whereas non-RGB modalities provide reduced visual exposure and can operate under different environmental conditions. [TODO: add references]

B. Privacy-Friendly and Non-RGB Human Perception

Privacy-friendly human perception aims to reduce reliance on raw visual appearance while preserving task-relevant information. Depth, mmWave, LiDAR, and WiFi-CSI are attractive in this context, but their signals are generally weaker or more unstable than RGB images. VK and skeleton representations reduce appearance information and preserve geometry, but they should not be treated as formal privacy guarantees. [TODO: add references]

C. Cross-Modal Knowledge Distillation and Privileged Information

Knowledge distillation and privileged information learning provide mechanisms for using strong training-time information to supervise a deployable model. RGB-to-pose or RGB-to-skeleton distillation is related to this paper. The distinction here is that RGB and VK are organized as a privileged structural supervision chain for non-RGB multimodal sensing, rather than as a direct RGB-to-skeleton student setting. [TODO: add references]

D. Missing-Modality Robust Multimodal Learning

Missing-modality learning has been addressed with modality dropout, reconstruction, shared latent spaces, and reliability-aware fusion. This paper does not claim missing-modality learning as a new problem. Instead, it incorporates modality-subset training into a privacy-friendly non-RGB human sensing pipeline supervised by RGB-VK privileged knowledge. [TODO: add references]

E. Human Pose Estimation and Human Activity Recognition

HPE and HAR represent complementary levels of human perception. HPE evaluates geometric body structure, while HAR evaluates semantic activity understanding. This paper uses both tasks to validate that the proposed sensing framework supports both structural and semantic perception in smart spaces. [TODO: add references]

III. Methodology

A. Problem Formulation

Let X^{rgb} denote the RGB observation available only during training. Let X^{vk} denote the visual keypoint representation derived from the RGB stream or associated pose annotations. The non-RGB modalities are denoted as X^d , X^l , X^m , and X^w , corresponding to depth, LiDAR, mmWave, and WiFi-CSI, respectively. For each sample, the available non-RGB modality set is $\mathcal{M} \subseteq \{d, l, m, w\}$, and its missing pattern is represented by a binary mask $\mathbf{m}$.

For HPE, the target is a 3-D human pose $Y^{hpe} \in \mathbb{R}^{J \times 3}$, where J is the number of joints. For HAR, the target is an activity label Y^{har} . During training, RGB, VK, and non-RGB sensors can be used to construct privileged supervision. During inference, RGB is not used, and the Student predicts HPE and HAR only from the available non-RGB modalities.

B. VK as Privacy-Friendly Structured Supervision

VK is used as a structured intermediate representation between raw RGB and non-RGB sensing. Unlike raw RGB, VK suppresses appearance, texture, clothing, and background details. Unlike purely non-RGB signals, it preserves body geometry and motion structure. Therefore, VK is suitable as a bridge for transferring RGB-derived structural knowledge to non-RGB sensors.

This paper does not claim that VK provides formal anonymity or cryptographic privacy. Instead, VK is treated as a privacy-friendly representation that reduces appearance-level visual exposure while preserving task-relevant body structure. This positioning is important for smart-space sensing, where the final deployed Student should not require RGB input.

C. RGB-VK Super-Teacher

The RGB-VK Super-Teacher is a training-time model that receives privileged RGB/VK information and learns a representation useful for both HPE and HAR. Its purpose is not to be a universal model that outperforms all task-specific teachers, but to organize visual privileged knowledge into a transferable supervision signal.

The Super-Teacher contains an RGB branch, a VK branch, a fusion module, and task heads for HPE and HAR. The RGB branch captures appearance and contextual cues, while the VK branch imposes geometric structure. Their fused representation provides privileged supervision to the non-RGB Student.

The Super-Teacher outputs teacher pose predictions, activity logits, and intermediate features. These outputs supervise the Student through task-level and representation-level distillation.

D. Non-RGB Multimodal Student

The Student receives only non-RGB modalities at inference. Each modality is encoded by a modality-specific

encoder and projected into a shared token dimension. The Student then produces modality-wise candidates and a fused representation for downstream HPE and HAR heads.

For HPE, the Student supports VK-assisted and non-visual settings in the current experiments. The non-visual setting uses depth, LiDAR, mmWave, and WiFi-CSI. For HAR, the current experiments use VK, depth, LiDAR, and mmWave, with a non-visual setting based on depth, LiDAR, and mmWave.

E. Reliability-Calibrated Fusion

For an available modality set $\mathcal{M}$, the Student estimates a reliability score α_i for each modality $i \in \mathcal{M}$. The fused prediction is computed as

$$\hat{Y} = \sum_{i \in \mathcal{M}} \alpha_i \hat{Y}_i, \quad \sum_{i \in \mathcal{M}} \alpha_i = 1, \quad \alpha_i \geq 0, \quad (1)$$

where $\hat{Y}_i$ is the candidate prediction from modality i . This design allows the model to adapt its decision weights according to the current modality combination and signal quality. [TODO: add reliability weight visualization]

F. Cross-Modal Distillation Objective

The training objective combines task supervision and distillation:

$$\mathcal{L} = \mathcal{L}_{hpe} + \lambda_{har} \mathcal{L}_{har} + \lambda_{kd} \mathcal{L}_{kd} + \lambda_{vk} \mathcal{L}_{vk} + \lambda_{rel} \mathcal{L}_{rel}. \quad (2)$$

For HPE, $\mathcal{L}_{hpe}$ measures pose regression error, and may include joint-level and bone-level losses. For HAR, $\mathcal{L}_{har}$ is the classification loss. $\mathcal{L}_{kd}$ distills teacher outputs or intermediate features into the Student. $\mathcal{L}_{vk}$ encourages alignment with VK-guided structural representations. $\mathcal{L}_{rel}$ regularizes reliability fusion. [TODO: verify final loss weights from code/config]

G. Missing-Modality Robust Training

During training, the Student randomly samples modality subsets and learns to predict under incomplete observations. This strategy prepares the model for arbitrary sensor availability at inference. This paper uses modality-subset training as a deployment-oriented robustness mechanism, not as a standalone novelty. Its role is to ensure that RGB-privileged knowledge can be used to train a non-RGB model that remains useful when some sensors fail.

H. Training and Inference Pipeline

Training proceeds in three stages. First, RGB and VK are used to train the Super-Teacher. Second, the non-RGB Student is trained with task supervision and teacher distillation under randomly sampled modality subsets. Third, the trained Student is evaluated under all modality combinations.

At inference, RGB is absent. The system receives only the available non-RGB sensors, estimates modality reliability, fuses modality-wise predictions, and outputs HPE or HAR results.

TABLE I
HPE main results on MMFi. Lower MPJPE is better.

Method	Scope	Avg. MPJPE (mm)	Full MPJPE (mm)
Full-modality teacher	VK+D+L+M+W	1594.89	48.45
Full-modality baseline	VK+D+L+M+W	617.17	49.14
Student-VK	VK+D+L+M+W	75.18	50.17
Student-NV	D+L+M+W	84.68	51.03

IV. Experiments

A. Dataset and Experimental Setup

Experiments are conducted on MMFi, which contains synchronized multimodal human sensing data. The HPE experiments use VK, depth, LiDAR, mmWave, and WiFi-CSI. The HAR experiments use VK, depth, LiDAR, and mmWave in the current implementation. The default protocol is protocol3 with random split unless otherwise specified.

[TODO: fill exact dataset split, number of subjects, environments, actions, and implementation details]

B. Evaluation Metrics

For HPE, we report mean per-joint position error (MPJPE), Procrustes-aligned MPJPE (PA-MPJPE), and mean squared error (MSE). MPJPE and PA-MPJPE are reported in millimeters. For HAR, we report classification accuracy and macro-F1. Missing-modality robustness is evaluated by averaging performance across all available modality combinations.

C. Main Results on HPE

Table ?? summarizes HPE results under random split. The full-modality teacher achieves strong performance when all modalities are available, but its average error across all missing-modality combinations is high. In contrast, the proposed Student-VK and Student-NV models maintain substantially lower average MPJPE across arbitrary combinations.

These results indicate that full-modality training alone is not sufficient for sensor-missing deployment. The Student trained with modality subsets provides much stronger robustness under arbitrary available modalities.

D. Main Results on HAR

Table ?? reports HAR results. The Student-VK achieves 85.15% average accuracy across 15 modality combinations and 96.56% full-modality accuracy. The non-visual Student also achieves strong full-modality accuracy without RGB.

The HAR results show that the proposed training strategy improves missing-modality robustness not only for geometric pose estimation but also for semantic activity understanding.

TABLE II
HAR main results on MMFi. Higher accuracy and macro-F1 are better.

Method	Scope	Avg. Acc. (%)	Full Acc. (%)	Full F1 (%)
Teacher	VK+D+L+M	66.25	95.57	95.56
Student-VK	VK+D+L+M	85.15	96.56	96.55
Student-NV	D+L+M	80.62	96.12	96.13
No distillation	VK+D+L+M	82.86	95.99	95.98
Uniform fusion	VK+D+L+M	67.46	95.39	95.39

TABLE III
Super-Teacher auxiliary results.

Model	Task	Avg. Result	Full Result
Super-Teacher	HPE	1538.81 mm	59.78 mm
Super-Teacher	HAR	53.59%	90.55%
Super-HPE Student-VK	HPE	82.06 mm	56.82 mm

TABLE IV
Ablation study. HPE uses MPJPE in mm and HAR uses accuracy in %.

Setting	Task	Avg.	Full
Student-VK	HPE	75.18	50.17
No distillation	HPE	72.68	48.83
Uniform fusion	HPE	82.77	52.28
Student-VK	HAR	85.15	96.56
No distillation	HAR	82.86	95.99
Uniform fusion	HAR	67.46	95.39

E. Effectiveness of RGB-VK Super-Teacher

The Super-Teacher is used as a training-time privileged knowledge organizer. Current results show that naive cross-task Super-Teacher evaluation is not consistently stronger than task-specific teachers. Therefore, the Super-Teacher is not claimed as a universally superior teacher. Instead, it supports the paper's training-time RGB-to-VK-to-non-RGB supervision chain.

These results should be discussed carefully. They motivate the RGB-VK privileged supervision idea but should not be used as the main performance evidence unless additional experiments show consistent benefit.

F. Ablation Study

Table ?? summarizes available ablation results. Reliability fusion improves over uniform fusion in both HPE and HAR. However, the HPE no-distillation ablation is currently strong, so additional output-KD combination evaluation is required before making strong claims about HPE distillation gains.

[TODO: add output-KD, w/o modality-subset training, and reliability weight visualization]

G. Missing-Modality Robustness

The HPE Student-VK is evaluated over 31 modality combinations, and the HAR Student-VK is evaluated over 15 combinations. The large gap between teacher average robustness and Student average robustness indicates that

explicitly training under modality subsets is essential for deployment.

[TODO: add grouped table by number of available modalities: single, two, three, four, full]

H. Generalization Results

The available cross-scene and cross-subject results provide evidence of protocol robustness. For HPE, Student-VK obtains 142.02 mm average MPJPE under cross-scene evaluation, and Student-NV obtains 153.89 mm. For HAR, Student-VK obtains 80.61% average accuracy under cross-scene and 85.30% under cross-subject.

[TODO: complete HPE cross-subject if available]

I. Privacy-Utility Analysis

The framework avoids RGB during inference. VK is used as a structured supervision signal that reduces exposure of visual appearance while preserving body geometry. This privacy-utility analysis should be supported by qualitative examples comparing RGB images and VK representations.

[TODO: add RGB vs VK visualization and privacy discussion figure]

J. Visualization and Case Study

Future visualization should include RGB and VK comparisons, HPE qualitative predictions under full and missing modalities, HAR confusion cases, reliability weights under different modality combinations, and failure cases under severe sensor degradation.

V. Discussion

VK works as a bridge because it retains geometric cues that are important to both HPE and HAR, while suppressing appearance details that make raw RGB privacy-sensitive. It is not a final inference modality in privacy-sensitive deployment; rather, it is an intermediate structured supervision signal.

Using RGB only during training allows the model to benefit from visual knowledge without requiring RGB cameras at deployment. This is particularly useful for smart spaces where camera installation may be socially or legally constrained.

HPE and HAR complement each other in the proposed framework. HPE evaluates whether the model learns body structure, while HAR evaluates whether this structure and multimodal evidence can support semantic behavior understanding.

The current study has limitations. First, VK is privacy-friendly but not a formal privacy guarantee. Second, some distillation ablations remain incomplete, especially for HPE. Third, Super-Teacher results should be interpreted as auxiliary evidence rather than definitive proof of cross-task teacher superiority.

VI. Conclusion

This paper presented an RGB-VK privileged distillation framework for privacy-friendly and missing-modality robust human sensing. The proposed framework uses RGB only as training-time privileged information, compresses visual knowledge through VK structured supervision, and trains a non-RGB multimodal Student for HPE and HAR. Experiments on MMFi show that the Student maintains strong performance under arbitrary modality combinations, supporting practical deployment in smart spaces where RGB inference is undesirable and sensor availability is unstable. Future work will extend the framework to broader real-world deployments, additional sensor configurations, and stronger privacy analysis.

References

- [1] [TODO: add MMFi dataset reference]
- [2] [TODO: add knowledge distillation and privileged information references]
- [3] [TODO: add missing-modality robust multimodal learning references]
- [4] [TODO: add multimodal human sensing references]
- [5] [TODO: add HPE and HAR references]